MILITARY STANDARD

MIL-STD-1320-151A (NAVY)

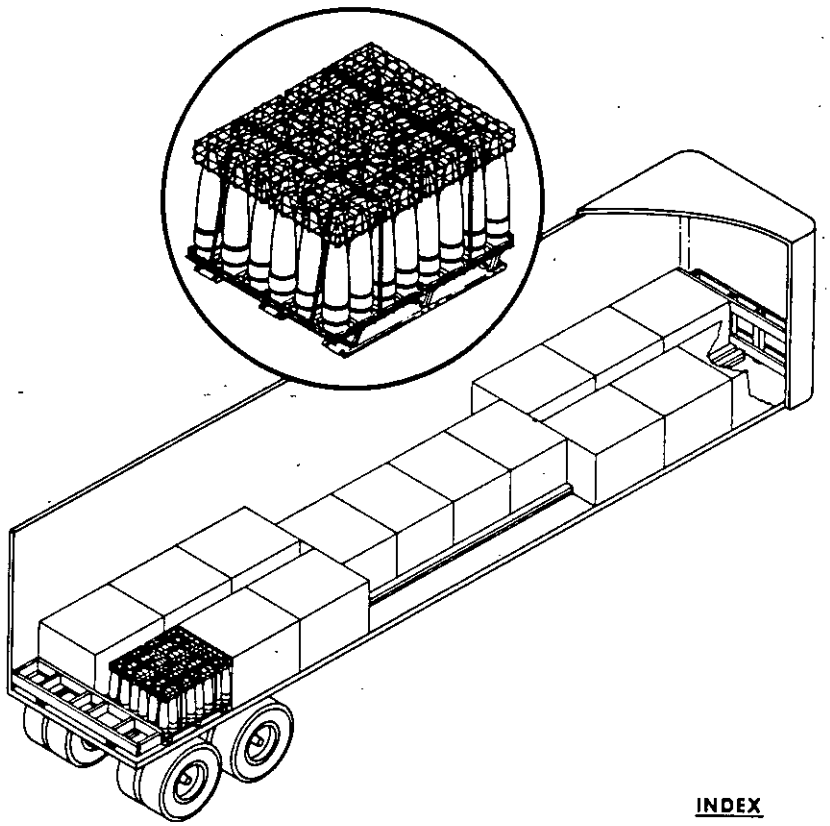
26 MAY 1976

SUPERSEDING
MIL-STD-1320-151 (NAVY)
1 MAY 1974

TRUCKLOADING
5"/38 PROJECTILE
IN PALLET ADAPTER MK 11
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/4
DIMENSIONS ..... SEE TABLE I
CUBE ..... SEE TABLE I
WEIGHT ..... SEE TABLE I
DOT HAZARD CLASS ..... SEE TABLE VI



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray NWHL — 5/20/76
TECHNICAL DIRECTION AGENT (TDA) — DATE

J E Kelly — 5/20/76
SEASYSCOM BY DIRECTION — DATE

ORIGINATOR L R Quinn — 5/20/76 DATE
NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF 5"/38 PROJECTILES WITH WATER-PROOF PROTECTIVE CAPS OR PROJECTILES WITH WINDSHIELDS IN PALLET ADAPTER MK 11 . THE UNIT LOADS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/4.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH ROLL UP DOORS.
3. LENGTH OF THE STRUT, PIECE 5 OF DETAIL B, SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
4. WHEN THE DISTANCE BETWEEN THE LADING AND TRAILER DOORS IS LESS THAN 12 INCHES, SUBSTITUTE REAR BLOCKING ASSEMBLY DETAIL C FOR DETAIL B. THE ASSEMBLY SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
5. WHEN TRAILERS HAVE NON-NAILABLE FLOORS, SUBSTITUTE FILLER ASSEMBLY, DETAIL D PAGE 4, FOR SLEEPERS. THE FILLER ASSEMBLY SHALL COMPLETELY FILL THE VOID PREVENTING LATERAL MOVEMENT.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER THE BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA. CLOSE AND SEAL TRAILER DOORS. ATTACH APPROPRIATE EXPLOSIVE PLACARDS FOR TYPE OF PROJECTIBLE BEING SHIPPED. (SEE TABLE VI, PAGE 4) TO BOTH SIDES, FRONT AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATION: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL & LTL 40, 42, 44, & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 14 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES II, III & IV ARE MET.

1. TABLES II, III & IV PROVIDE THE CORRECT LOAD PATTERN FOR THE LENGTH AND WEIGHT OF THE TRAILER BEING LOADED, PERMISSABLE GROSS AXLE AND GROSS VEHICLE WEIGHTS, AND THE NUMBER OF UNIT LOADS BEING LOADED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILER: THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7200 LBS AND OF THE DRIVE TANDEM AXLES 10,500 LBS.
STANDARD WEIGHT TRACTOR-TRAILER: THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7800 LBS AND OF THE DRIVE TANDEM AXLES 11,500 LBS.

2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR LENGTH OF TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT OF THE TRACTOR-TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED.
3. BLOCK AND BRACE THE SELECTED LOAD PATTERN USING THE PRINCIPLES OF THE TRUCKLOADS SHOWN ON PAGES 5 AND 6.
4. THE TRUCKLOADS SHOWN ON PAGES 5 AND 6 DOES NOT LIMIT SHIPMENT TO THE NUMBER OF UNIT LOADS SHOWN. THEY ARE PRESENTED TO SHOW THE BASIC PRINCIPLES OF LOADING.
5. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN TABLES II, III & IV, FOLLOW THE PRINCIPLES OF THE LOAD SHOWN ON PAGE 6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

TABLE I

	UNIT LOAD DIMENSIONS	CUBE (CU FT)	UNIT LOAD WT PALLET ADAPTER MK 11 MOD I (LBS)
PROJECTILE WITH WATER-PROOF CAP	40L X 48W X 28 3/8H	31.6	3030
PROJECTILE WITH WIND-SHIELD	40L X 48W X 27 1/2H	30.5	2910

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TABLE II

40 FT TRAILER

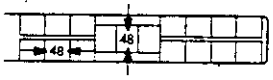
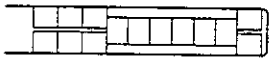
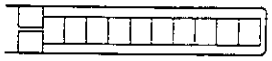
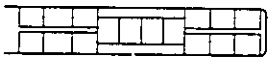
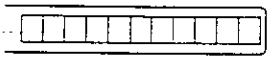
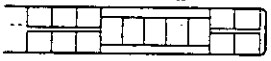
NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
17***		14*		12*	
16***		13*		11*	
15**					

TABLE III

42 FT TRAILER

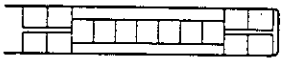
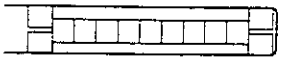
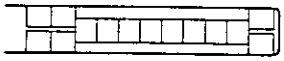
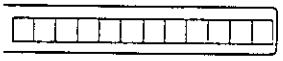
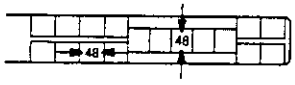
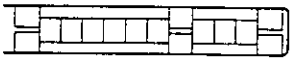
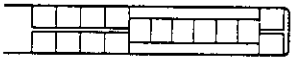
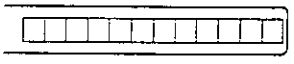
NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
17***		15**		13*	
16***		14*		12*	

TABLE IV

44/45 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
17***		15**		13*	
16***		14*		12*	

*** AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILER ONLY WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS.

** AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THORUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHT OF 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHTS OF 80,000 LBS.

* AUTHORIZED FOR STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 32,000 LBS ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 73,280 LBS.

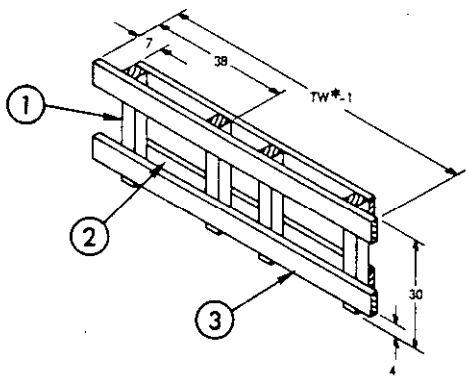
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TABLE V
TRUCKLOAD WEIGHTS

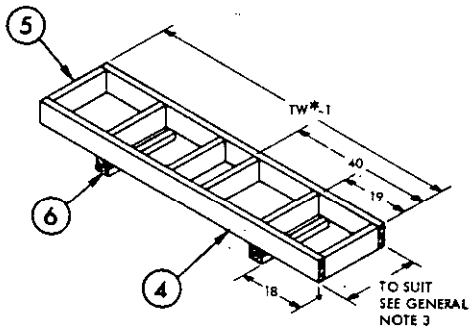
NUMBER OF UNIT LOADS	TRUCKLOAD WT (LBS)	
	PROJ W/WATER-PROOF CAP	PROJ W/ WINDSHIELD
17	TOO HEAVY	49,470
16	48,480	46,560
15	45,450	43,650
14	42,420	40,740
13	39,390	37,830
12	36,360	34,920

TABLE VI

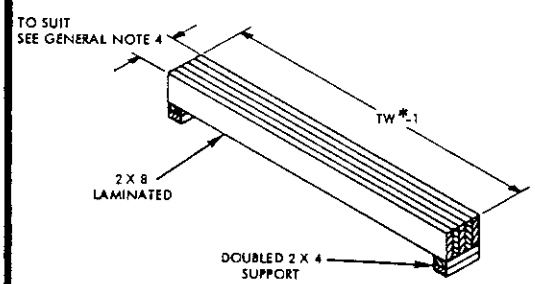
HAZARD CLASSIFICATION	
TYPE OF PROJECTILES	DOT
AAC	EXPLOSIVES A
COM	EXPLOSIVES A
FCLVT	EXPLOSIVES A
FCLVTNF	EXPLOSIVES B
FCLPDNF	EXPLOSIVES B
HC	EXPLOSIVES A
ILLUM	EXPLOSIVES B
VTNF	EXPLOSIVES B
VT	EXPLOSIVES A
WP	EXPLOSIVES A



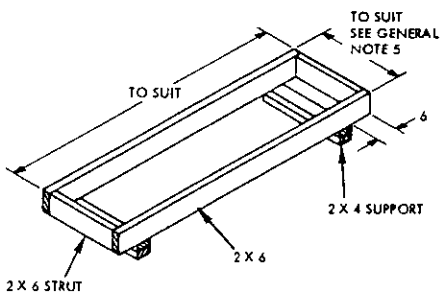
DETAIL A
FRONT BULKHEAD
1 REQUIRED



DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED



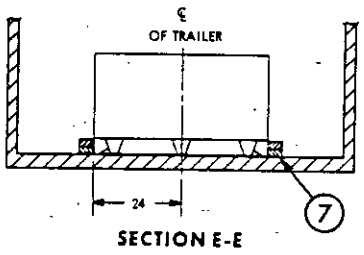
DETAIL C
REAR BLOCKING ASSEMBLY
SEE GENERAL NOTE 4



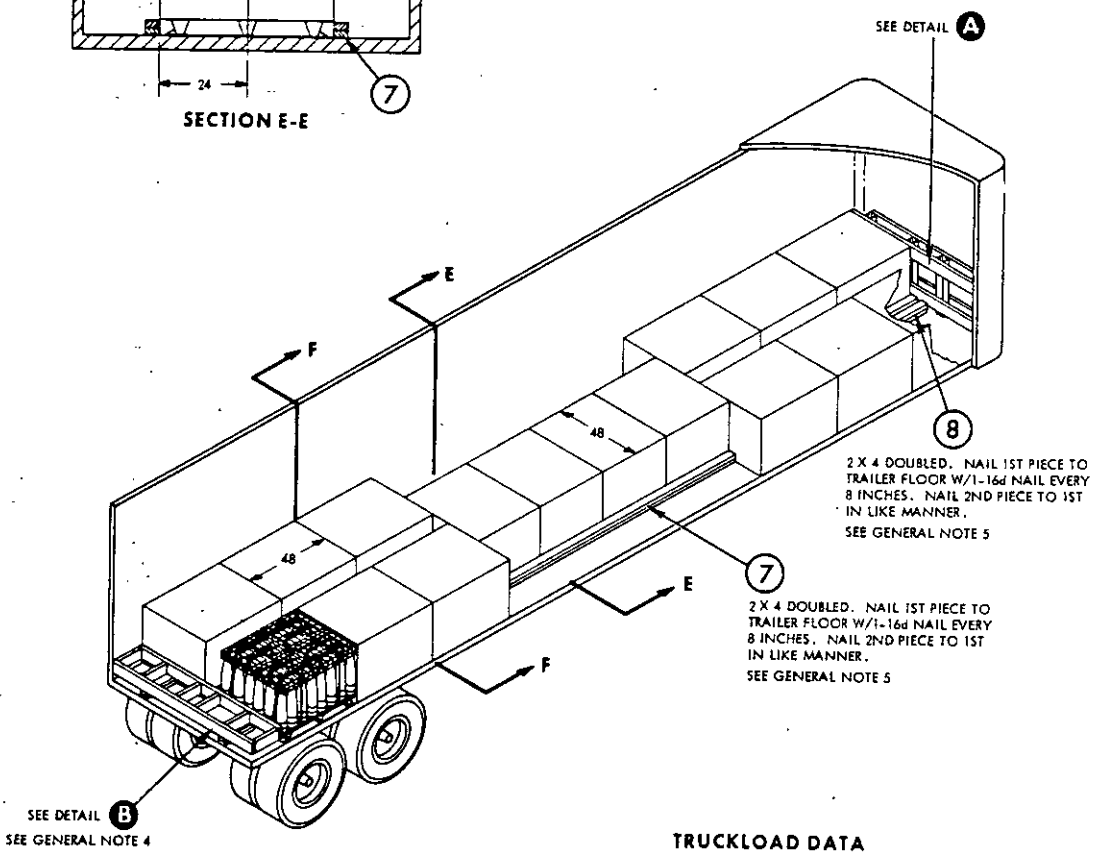
DETAIL D
FILLER ASSEMBLY
FOR TRAILERS WITH
NON-NAILABLE FLOORS

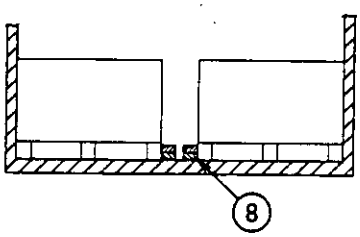
* TW=TRAILER WIDTH

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SECTION E-E





SECTION F-F

TRUCKLOAD DATA

* TW = TRAILER WIDTH

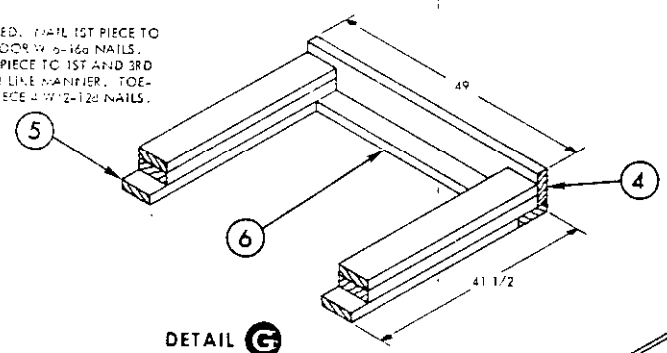
SEE TABLE

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
8	SLEEPER	2 X 4 X 48	24	SEE FIELD NOTE		
7	SLEEPER	2 X 4 X 40	16	SEE FIELD NOTE		
6	SUPPORT	2 X 4 TO SUIT	4	4	2 PER JOINT	12d
5	STRUT	2 X 6 TO SUIT	6	SEE 4	–	–
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3 PER JOINT	16d
3	AFT CROSSMEMBER	2 X 6 X TW*-1	2	1	3 PER JOINT	12d
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	1	3 PER JOINT	12d
1	VERTICAL	4 X 4 X 30	4	SEE 2 & 3	–	–

LIST OF MATERIALS & NAILING DATA

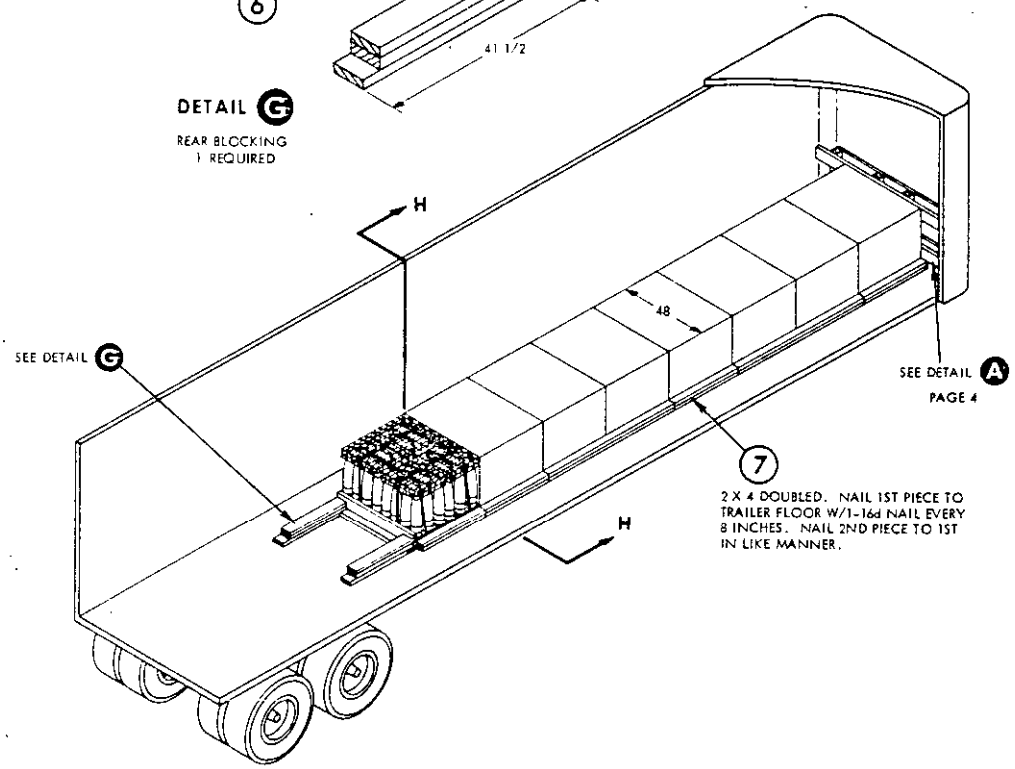
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2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/ 6-16d NAILS. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER. TOE-NAIL TO PIECE 4 W/2-12d NAILS.

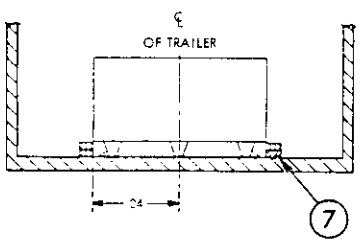


DETAIL G

REAR BLOCKING
1 REQUIRED



2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



SECTION H-H

* TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
7	SLEEPER	2 X 4 X RANDOM LENGTHS	107 LIN FT	SEE FIELD NOTE		
6	LOWER CROSSMEMBER	2 X 6 X 49	1	1	6	12d
5	BACKUP CLEAT	2 X 6 X 36	6	SEE DETAIL G		
4	UPPER CROSSMEMBER	2 X 6 X 49	1	-	-	-
3	AFT CROSSMEMBER	2 X 6 X TW* - 1	2	1	3 PER JOINT	12d
2	FWD CROSSMEMBER	2 X 6 X TO SUIT	2	1	3 PER JOINT	12d
1	VERTICAL	4 X 4 X 30	4	SEE 2 & 3	-	-

LIST OF MATERIALS & NAILING DATA

REVIEW ACTIVITIES
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N335

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